

# **SYNOPSIS OF EXPLORATORY PROJECT (EP)**

**ON**

## **OptiChoice:** **Intelligent Decision Evaluation System**

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**Submitted by:**

Chirayu Mitra – 2310991558

Rhythm Gaur – 2310992349

Yash Vardhan Singh– 2310992485

Shubhank Gaur – 2310992530

**Supervised By:**

Dr. Preeti Saini

Assistant Professor

Chitkara University



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY**  
**CHITKARA UNIVERSITY, PUNJAB, INDIA**

# **CONTENTS**

| <b>S. No.</b> | <b>Title</b>                            | <b>Page No.</b> |
|---------------|-----------------------------------------|-----------------|
| 1             | Abstract                                | 3               |
| 2             | 2. Introduction                         | 4               |
|               | 2.1 Background and Context              | 4               |
|               | 2.2 Problem Statement                   | 5               |
|               | 2.3 Significance of the Project         | 6               |
|               | 2.4 Objectives                          | 7               |
|               | 2.5 Scope and Limitations               | 7               |
| 3             | 3. Methodology                          | 8               |
|               | 3.1 Technical Approach                  | 8               |
|               | 3.2 System Architecture                 | 9               |
|               | 3.3 Functional Modules Overview         | 10              |
|               | 3.4 Workflow Diagram                    | 11              |
|               | 3.5 Transparent Hybrid Decision Model   | 12              |
| 4             | 4. Tools and Technologies               | 13              |
|               | 4.1 Programming Languages               | 13              |
|               | 4.2 Frameworks / Libraries              | 13              |
|               | 4.3 Databases / Storage                 | 13              |
|               | 4.4 IDEs / Platforms                    | 13              |
|               | 4.5 APIs / External Integrations        | 13              |
|               | 4.6 Hosting / Deployment Tools          | 13              |
| 5             | 5. Project Plan                         | 14              |
|               | 5.1 Module-wise Feature Breakdown       | 14              |
|               | 5.2 Week-wise Development Timeline      | 15              |
|               | 5.3 Expected Outcomes                   | 16              |
|               | 5.4 Risk Factors / Assumptions          | 16              |
| 6             | 6. References / Literature Review Table | 17              |
|               | References (APA Style)                  | 18              |

## 1. Abstract

The increasing complexity of academic, professional, and personal decision-making has created a need for intelligent systems that can help users evaluate multiple alternatives based on multiple factors. Traditional methods such as manual comparison, pros-and-cons lists, and spreadsheets often lack personalization, structured evaluation, and the ability to analyse how different priorities affect the final decision.

To address these challenges, **AI Decision Engine** is proposed as an intelligent web-based decision-support system that helps users analyse complex choices and identify suitable alternatives based on their priorities. The application allows users to define a decision problem, add multiple alternatives, create evaluation criteria, assign importance weights, and provide ratings or values for each alternative. A **multi-criteria decision-making engine** processes this information to calculate weighted scores and generate a transparent ranking of the available options.

The system also integrates **Generative AI** to provide contextual analysis and natural-language explanations. The AI component analyses the decision context and calculated results to identify key trade-offs, strengths, weaknesses, and important factors influencing the recommendation. The platform also includes a “**What-If**” **scenario analysis** feature, allowing users to modify criteria weights or assumptions and observe how these changes affect the ranking.

The proposed frontend will be developed using **React.js and TypeScript**, while **Node.js and Express.js** will be used to build RESTful APIs and manage application logic. **PostgreSQL**, integrated through **Supabase**, will be used for storing users, decisions, alternatives, criteria, scenarios, and recommendation history. A Generative AI API will be securely integrated through server-side services.

**AI Decision Engine** demonstrates the integration of Artificial Intelligence, multi-criteria decision-making, explainable recommendations, and full-stack web development to provide a transparent, personalized, and interactive decision-support solution.

## **2. Introduction**

### **2.1 Background and Context**

Decision-making is an essential part of academic, professional, and personal life. Students compare career paths and technologies, job seekers evaluate multiple offers, teams prioritize projects, and individuals regularly choose between alternatives that cannot be judged using a single factor. A higher salary, for example, may come with greater relocation cost or risk; a technically interesting project may require more time; and an option that performs well on one criterion may perform poorly on another. The difficulty therefore lies not only in collecting information, but in balancing several competing criteria according to what matters most to the individual making the decision.

Traditional decision-making is frequently informal. Users may create simple pros-and-cons lists, compare options mentally, or rely on generic advice from search engines and conversational AI tools. Such approaches can be useful, but they do not always preserve a consistent evaluation structure. A generic recommendation may not explicitly record the user's priorities, show the numerical contribution of each criterion, or make it easy to test whether the conclusion changes when the user's priorities change.

The proposed AI Decision Engine addresses this gap by combining a structured multi-criteria decision model with AI-assisted contextual analysis. The structured layer captures alternatives, criteria, ratings, and importance weights. It calculates a transparent score for every alternative and produces a ranking that can be inspected by the user. The AI layer does not replace this calculation; instead, it interprets the decision context, explains major trade-offs, summarizes qualitative considerations, and presents the results in natural language. This hybrid design is intended to provide both computational consistency and user-friendly intelligence.

## **2.2 Problem Statement**

Many real-world decisions involve multiple alternatives and criteria that are difficult to compare simultaneously. Users may know their available options but remain uncertain about how to balance factors such as cost, growth potential, convenience, learning value, risk, location, time commitment, or long-term benefit. When these factors conflict, an unstructured comparison can lead to inconsistent reasoning, overlooked assumptions, or decisions based primarily on whichever factor appears most prominent at the moment.

Existing AI assistants can generate advice from natural-language prompts, but a conversational response alone may not provide a repeatable decision model. It may be difficult for the user to verify which priorities influenced the answer, compare several options side by side, or understand how a small change in assumptions would affect the recommendation. Similarly, conventional comparison tables may provide numerical scores but lack contextual interpretation.

Therefore, the core problem addressed by this project is the absence of an accessible, transparent, and personalized system that combines explicit multi-criteria evaluation with AI-generated insight. The proposed system will enable users to define their own decision criteria and priorities, calculate and inspect rankings, analyze trade-offs, and perform scenario-based “What-If” comparisons before making a final decision.

## **2.3 Significance of the Project**

The project is significant because it transforms an informal and often subjective activity into a structured decision-support workflow without removing user control. Instead of assuming that every user values the same outcome, the system allows the importance of each criterion to be defined by the user. The resulting recommendation is therefore based on an explicit representation of priorities rather than a fixed rule.

The project also emphasizes explainability. The deterministic scoring component can show how criteria and weights contribute to the ranking, while the AI component can convert these results into understandable explanations. This is particularly useful when an option wins overall but has weaknesses in important areas. The user can inspect the reasons, modify assumptions, and test alternative scenarios rather than accepting a single answer as final.

From an academic perspective, the project combines several current areas of Computer Science: web application engineering, data modeling, algorithmic multi-criteria evaluation, Generative AI integration, explainable AI interaction, visualization, authentication, and cloud deployment. The same core framework can be adapted to multiple non-regulated domains such as career planning, technology selection, academic choices, project prioritization, consumer comparisons, and productivity planning. This broad applicability also provides meaningful future scope beyond the beta version.

## **2.4 Objectives**

- To develop an intelligent web-based platform for creating and comparing multi-option decisions using user-defined criteria.
- To implement a transparent multi-criteria scoring and ranking mechanism based on normalized criterion values and user-assigned importance weights.
- To integrate Generative AI for contextual interpretation, trade-off analysis, and explainable natural-language recommendations without replacing deterministic scoring.
- To provide interactive “What-If” scenario analysis so that users can examine the impact of changing priorities or assumptions.
- To deploy a beta version with decision history, visual analytics, report generation, and documentation suitable for copyright-oriented project delivery.

## **2.5 Scope and Limitations**

The beta version will provide a general-purpose decision workspace for personal, academic, and professional scenarios in which the user can explicitly define alternatives and evaluation criteria. Users will be able to create a decision, add multiple options, assign ratings or values, select criteria, set importance weights, generate a transparent ranking, obtain AI-assisted analysis, perform “What-If” simulations, save decisions, and export a structured report.

The project will focus on decisions where the user is able to provide or estimate the information required for comparison. The initial implementation will support guided manual entry rather than automatically collecting verified external data from every possible source. The system will also distinguish between deterministic scoring and AI-generated interpretation so that the ranking logic remains inspectable.

The beta version will not claim to predict future outcomes or guarantee that a recommendation is objectively correct. Results will depend on the completeness and quality of user inputs, the selected criteria, the assigned weights, and the assumptions used during evaluation. High-stakes decisions that require regulated professional judgement, including medical diagnosis, legal advice, or other domains where automated recommendations could create unacceptable risk, are outside the intended scope of the project.

### **3. Methodology**

#### **3.1 Technical Approach**

The proposed solution follows a client-server and service-oriented architecture. The frontend provides an interactive workspace where users define a decision problem, enter alternatives, configure criteria, assign weights, and view results. The backend exposes secure APIs, validates inputs, applies normalization where required, executes the decision-scoring algorithm, stores persistent data, and coordinates requests to the AI analysis service.

The core recommendation logic is intentionally hybrid. First, a deterministic multi-criteria evaluation layer converts each alternative into comparable criterion values. For each alternative, the system computes an overall score using the normalized criterion value and the importance weight selected by the user. A representative formulation is:

$$\text{Score}(A_j) = \sum [w_i \times v_{ij}]$$

where  $w_i$  represents the normalized importance of criterion  $i$  and  $v_{ij}$  represents the normalized performance of alternative  $j$  on that criterion. This layer provides a reproducible ranking and enables the system to display a score breakdown rather than only a final recommendation.

Second, a Generative AI service receives carefully structured decision context, the calculated ranking, relevant criteria, and qualitative notes supplied by the user. It is used to identify notable trade-offs, explain why alternatives differ, highlight assumptions, and produce a concise natural-language interpretation. The AI service is not permitted to silently replace the deterministic ranking; instead, its role is to augment the structured analysis with contextual explanation.

The “What-If” simulator will allow the user to modify weights, ratings, or assumptions and trigger recalculation. Multiple scenarios can be saved and compared. This design makes the system suitable for sensitivity analysis: if a small change in one criterion causes a large change in ranking, the platform can highlight that the decision is sensitive to that priority.

### **3.2 System Architecture**

The proposed architecture is divided into clearly separated layers to improve maintainability and testability. The Presentation Layer contains the web interface and visualization components. The Application Layer manages authentication, validation, API routing, and orchestration. The Decision Intelligence Layer performs normalization, weighted scoring, ranking, and sensitivity calculations. The AI Analysis Layer uses structured inputs to generate contextual explanations and trade-off analysis. Persistent data is maintained in PostgreSQL/Supabase, while external services are isolated behind backend APIs so that API keys and model credentials are not exposed in the client application.

The principal data entities will include User, Decision, Alternative, Criterion, CriterionWeight, AlternativeEvaluation, Scenario, RecommendationSnapshot, and Report. A Decision is associated with multiple Alternatives and Criteria; each AlternativeEvaluation links an alternative to a criterion-specific value. Scenario records allow the same decision to be recalculated under different assumptions. This relational structure preserves decision history and makes it possible to reconstruct why a particular ranking was generated.

**AI Decision Engine — Proposed System Architecture**

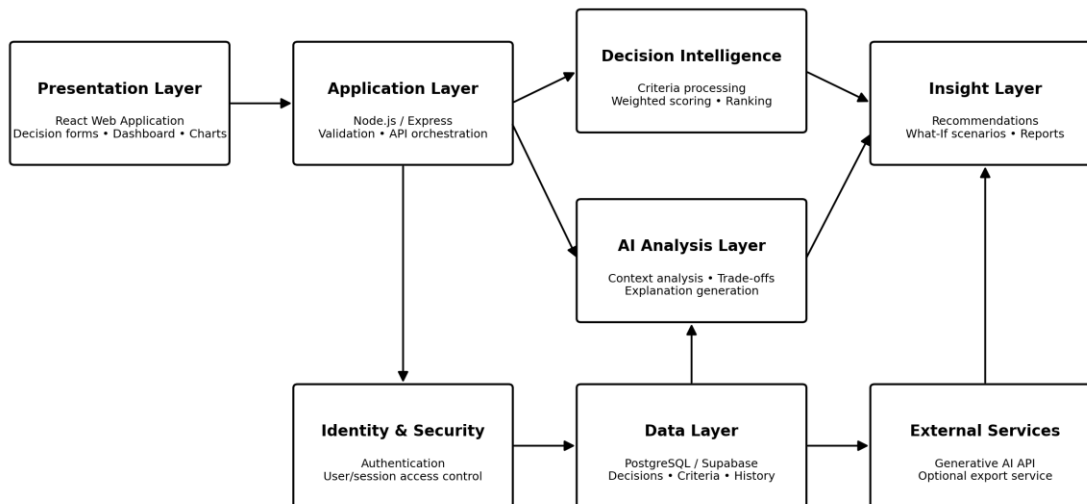


Figure 1. Layered System Architecture of the Proposed AI Decision Engine

### **3.3 Functional Modules Overview**

- 1. Authentication and User Profile:** Provides secure account creation, login, session handling, and a personal workspace. The profile is used to associate decisions, scenarios, reports, and preference history with the correct user.
- 2. Decision Workspace and Alternative Management:** Allows users to define the decision question, provide contextual notes, create multiple alternatives, edit option details, duplicate an existing decision, and organize decisions by category or status.
- 3. Criteria Definition and Priority Weighting:** Enables users to add custom evaluation criteria such as cost, growth, risk, location, learning value, or convenience. Users assign importance weights, and the system validates and normalizes those weights before scoring.
- 4. Alternative Evaluation and Data Normalization:** Collects criterion-specific ratings or values for each alternative. Where different scales are used, the module applies appropriate normalization so that criteria can be combined in a comparable manner.
- 5. Multi-Criteria Scoring and Ranking Engine:** Calculates weighted scores, produces an ordered ranking, and exposes a criterion-level score breakdown. This deterministic module is the transparent computational foundation of the recommendation.
- 6. AI Context and Trade-Off Analysis:** Uses a Generative AI service to interpret qualitative notes and structured results. It summarizes strengths and weaknesses, explains trade-offs, identifies potentially important assumptions, and generates readable decision insights.
- 7. What-If Scenario and Sensitivity Simulator:** Allows users to change criterion weights, option values, or assumptions and immediately recalculate the ranking. Scenario comparison highlights how stable or sensitive a recommendation is.
- 8. Explainable Recommendation Dashboard:** Presents rankings, charts, score contributions, key drivers, risks, pros and cons, and AI-generated explanations in a consolidated and easy-to-read interface.
- 9. Decision History, Reports and Export:** Stores previous decisions and scenarios, allows users to revisit or compare them, and generates a structured PDF summary containing the alternatives, criteria, weights, scores, recommendation, and assumptions.
- 10. Validation, Feedback and Audit Support:** Records important calculation inputs and validation results so that a user can understand the basis of a recommendation. This module also supports error handling and future feedback mechanisms.

### **3.4 Workflow Diagram**

The workflow begins when a user creates a decision and states the objective. The user then enters the alternatives that are to be compared and defines the criteria that will be used for evaluation. Each criterion receives an importance weight, and the user provides ratings, values, or structured notes for each alternative. The backend validates the information, normalizes comparable values where required, and calculates a weighted score for every alternative.

After ranking is complete, the AI analysis layer receives the structured results and relevant contextual information. It generates an explanation of the main factors driving the ranking, identifies strengths and weaknesses, and highlights important trade-offs. The user can then modify priorities or assumptions through the “What-If” module. Each scenario can be recalculated, compared with the original decision, saved to history, and exported as a report.

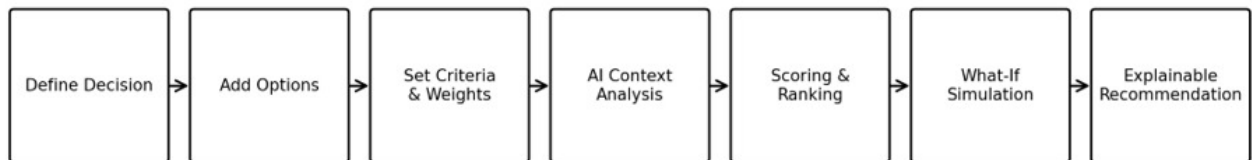


Figure 2. End-to-End Workflow from Decision Definition to Explainable Recommendation

### **3.5 Transparent Hybrid Decision Model**

A key design principle of the proposed system is separation between the calculation of a ranking and the generation of a natural-language explanation. The scoring engine uses explicit user-defined inputs and reproducible calculations. The AI layer receives the structured result and supporting context after scoring and is responsible for interpretation, not hidden replacement of the ranking. This approach supports traceability because the user can inspect the criteria, weights, option values, calculated score, and resulting order before reading the AI explanation.

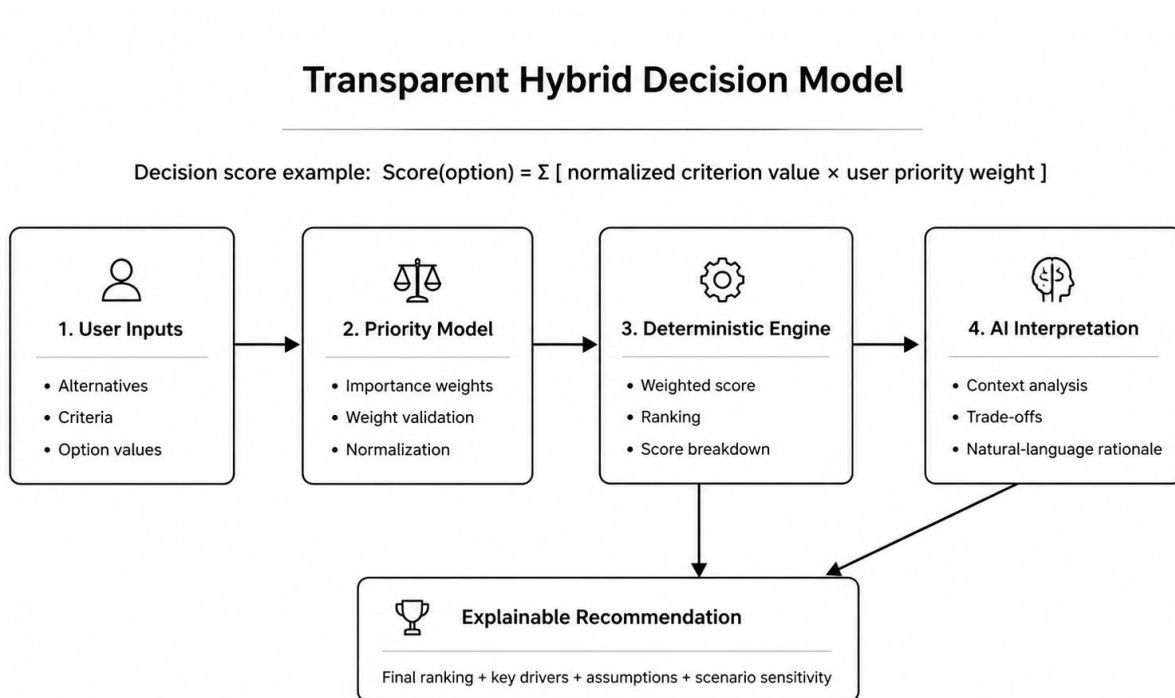


Figure 3. Hybrid Model Combining Deterministic Scoring with AI-Based Interpretation

## **4. Tools and Technologies**

### **4.1 Programming Languages**

JavaScript/TypeScript will be used for application logic and frontend-backend development, while HTML and CSS will support document structure and presentation. The final implementation may use TypeScript for stronger type safety in decision and scoring models.

### **4.2 Frameworks / Libraries**

React will be used for the interactive user interface; Node.js and Express.js will support server-side APIs. A UI styling framework such as Tailwind CSS may be used for responsive layouts, while a charting library such as Chart.js or Recharts can visualize rankings and scenario comparisons.

### **4.3 Databases / Storage**

PostgreSQL, accessed through Supabase or another managed PostgreSQL service, will store users, decisions, alternatives, criteria, evaluations, scenarios, and recommendation history. The relational model is suitable because the core entities have clear one-to-many and many-to-one relationships.

### **4.4 IDEs / Platforms**

Visual Studio Code will be used as the primary development environment. GitHub will support version control and collaboration, while Postman or an equivalent API client will be used for endpoint testing.

### **4.5 APIs / External Integrations**

A Generative AI API will provide contextual analysis and natural-language explanations. The API will be invoked only from secure backend or server-side functions so that credentials are not exposed in the browser. A PDF generation utility will support decision report export.

### **4.6 Hosting / Deployment Tools**

The frontend can be deployed using Vercel, while backend services can be deployed using a suitable cloud service such as Render, Railway, or serverless/edge functions. Supabase will provide managed authentication and PostgreSQL storage where selected.

## **5. Project Plan**

### **5.1 Module-wise Feature Breakdown**

| <b>Module</b>                                 | <b>Detailed Description</b>                                                                                                                                                                                                              |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication and User Profile               | Provides secure account creation, login, session handling, and a personal workspace. The profile is used to associate decisions, scenarios, reports, and preference history with the correct user.                                       |
| Decision Workspace and Alternative Management | Allows users to define the decision question, provide contextual notes, create multiple alternatives, edit option details, duplicate an existing decision, and organize decisions by category or status.                                 |
| Criteria Definition and Priority Weighting    | Enables users to add custom evaluation criteria such as cost, growth, risk, location, learning value, or convenience. Users assign importance weights, and the system validates and normalizes those weights before scoring.             |
| Alternative Evaluation and Data Normalization | Collects criterion-specific ratings or values for each alternative. Where different scales are used, the module applies appropriate normalization so that criteria can be combined in a comparable manner.                               |
| Multi-Criteria Scoring and Ranking Engine     | Calculates weighted scores, produces an ordered ranking, and exposes a criterion-level score breakdown. This deterministic module is the transparent computational foundation of the recommendation.                                     |
| AI Context and Trade-Off Analysis             | Uses a Generative AI service to interpret qualitative notes and structured results. It summarizes strengths and weaknesses, explains trade-offs, identifies potentially important assumptions, and generates readable decision insights. |
| What-If Scenario and Sensitivity Simulator    | Allows users to change criterion weights, option values, or assumptions and immediately recalculate the ranking. Scenario comparison highlights how stable or sensitive a recommendation is.                                             |
| Explainable Recommendation Dashboard          | Presents rankings, charts, score contributions, key drivers, risks, pros and cons, and AI-generated explanations in a consolidated and easy-to-read interface.                                                                           |
| Decision History, Reports and Export          | Stores previous decisions and scenarios, allows users to revisit or compare them, and generates a structured PDF summary containing the alternatives, criteria, weights, scores, recommendation, and assumptions.                        |
| Validation, Feedback and Audit Support        | Records important calculation inputs and validation results so that a user can understand the basis of a recommendation. This module also supports error handling and future feedback mechanisms.                                        |

## **5.2 Week-wise Development Timeline**

| <b>Module</b>                                  | <b>Planned Weeks</b> | <b>Expected Deliverables</b>                                                                    |
|------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------|
| Module 1: Requirements, Research & UI Planning | Week 1–2             | Use-case definition, decision scenarios, wireframes, database entities and acceptance criteria. |
| Module 2: Authentication & Core Data Model     | Week 2–3             | User authentication, database schema, secure CRUD foundation.                                   |
| Module 3: Decision Workspace & Alternatives    | Week 3–4             | Decision creation, alternative management and contextual input.                                 |
| Module 4: Criteria, Weighting & Evaluation     | Week 4–5             | Criteria management, weight validation and option evaluation forms.                             |
| Module 5: Scoring, Normalization & Ranking     | Week 5–6             | Working deterministic decision engine with score breakdowns.                                    |
| Module 6: AI Context & Explanation Layer       | Week 6–7             | Secure AI integration, trade-off analysis and explanation outputs.                              |
| Module 7: What-If & Sensitivity Analysis       | Week 7–8             | Scenario creation, recalculation and side-by-side comparison.                                   |
| Module 8: Dashboard, History & Reports         | Week 8–9             | Charts, decision history and PDF report export.                                                 |
| Testing, Security & Deployment                 | Week 9–10            | Integration testing, validation, deployment and beta stabilization.                             |
| Final Documentation & Report                   | Week 11              | Final formatted documentation, diagrams, references and copyright preparation.                  |
| Final Demo + Evaluation                        | Week 12              | Deployed demonstration, presentation, code freeze and evaluation.                               |

### **5.3 Expected Outcomes**

- Deployable beta version of an AI-assisted, multi-criteria decision-support web application.
- Personalized comparison of alternatives using user-defined criteria and importance weights.
- Transparent weighted scoring, normalization, ranking, and criterion-level score breakdown.
- AI-generated contextual insights that explain strengths, weaknesses, trade-offs, and important assumptions.
- Interactive “What-If” analysis and scenario comparison to test how changing priorities affects results.
- Persistent decision history and professionally generated decision reports.
- Original software documentation and implementation suitable for preparation toward copyright filing as the selected project outcome.

### **5.4 Risk Factors / Assumptions**

- AI-generated explanations may occasionally be inconsistent or overconfident. Structured prompts, bounded input context, validation, and visible score breakdowns will reduce dependence on unsupported AI claims.
- The quality of a recommendation depends on user-supplied ratings and priorities. The interface will clearly communicate that the system is a decision-support aid and not a guarantee of a correct real-world outcome.
- Different criteria may use different units or scales. Normalization rules and guided input formats will be implemented to reduce invalid comparisons.
- External AI APIs may introduce rate limits, latency, or cost. The deterministic scoring and ranking engine will remain functional even if AI analysis is temporarily unavailable.
- Scenario complexity can increase quickly when many alternatives and criteria are added. The beta version will define practical limits and optimize calculations for the intended user scale.

## 6. References / Literature Review Table

| S. No. | Source Title / Tool                                                                                                      | Key Idea                                                                               | Project Relevance                                                                      |
|--------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1      | Saaty, T. L. (1980). The Analytic Hierarchy Process.                                                                     | Presents a structured method for comparing alternatives using criteria and priorities. | Provides theoretical grounding for explicit weighting and multi-criteria comparison.   |
| 2      | Keeney, R. L., & Raiffa, H. (1993). Decisions with Multiple Objectives.                                                  | Explains value trade-offs and structured analysis of conflicting objectives.           | Supports the project's treatment of competing priorities and trade-offs.               |
| 3      | Shrestha, Y. R., Ben-Menahem, S. M., & von Krogh, G. (2019). Organizational decision-making structures in the age of AI. | Discusses AI as an augmentation mechanism within human decision processes.             | Supports the hybrid design in which users retain control over criteria and priorities. |
| 4      | Amershi, S., et al. (2019). Guidelines for Human-AI Interaction.                                                         | Provides principles for predictable, understandable, and controllable AI interactions. | Guides explainability, feedback, transparency, and user control.                       |
| 5      | NIST (2023). AI Risk Management Framework 1.0.                                                                           | Provides a framework for considering trustworthy and responsible AI risks.             | Supports risk discussion and responsible framing of AI-generated recommendations.      |
| 6      | Tervonen, T., & Figueira, J. R. (2008). Survey on stochastic multicriteria acceptability analysis methods.               | Reviews multi-criteria approaches and uncertainty considerations.                      | Informs possible future extensions involving uncertainty and robustness analysis.      |
| 7      | React Documentation.                                                                                                     | Describes component-based frontend development and state-driven interfaces.            | Supports implementation of the interactive decision workspace and dashboard.           |
| 8      | Node.js Documentation.                                                                                                   | Describes JavaScript runtime capabilities for server-side applications.                | Supports backend API, scoring orchestration, and secure service integration.           |
| 9      | Supabase Documentation.                                                                                                  | Documents authentication and managed PostgreSQL services.                              | Supports the proposed persistent data model and user authentication architecture.      |
| 10     | Generative AI API Documentation (selected provider).                                                                     | Documents structured inference and API integration for language models.                | Supports AI-based context analysis and natural-language recommendation explanations.   |

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